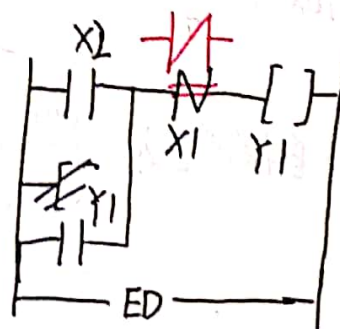
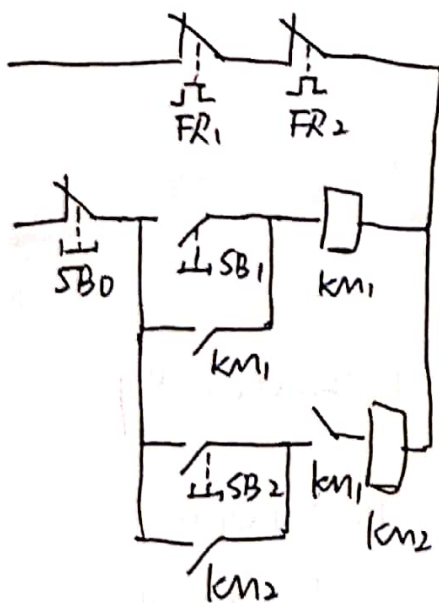
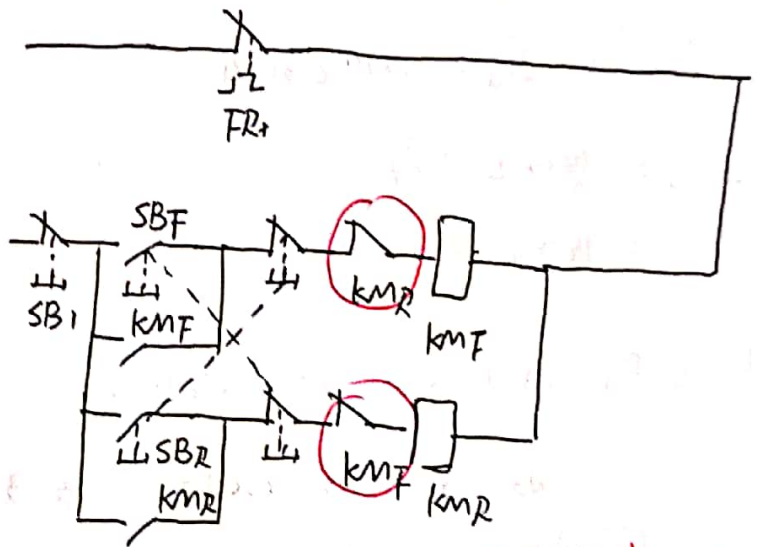
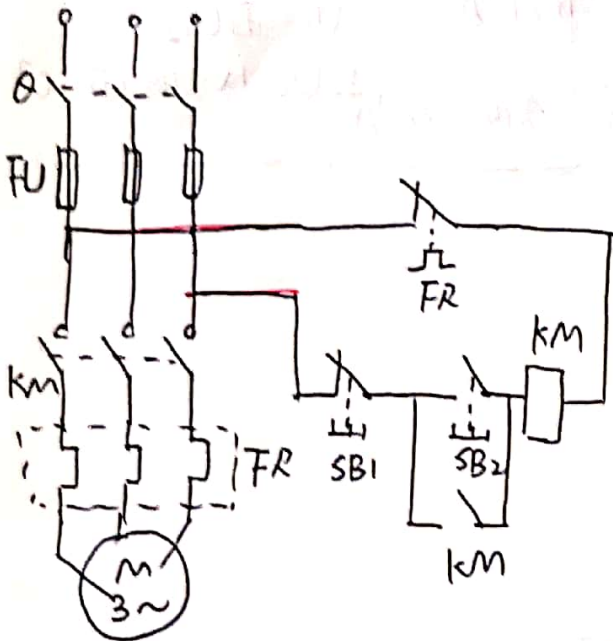


继电器接触控制电路分析

热继电器用来保护过载 FR

熔断器 短路保护 FU

空气断路器 用于短路、过载和欠压保护 KM



OR 并, AND 串

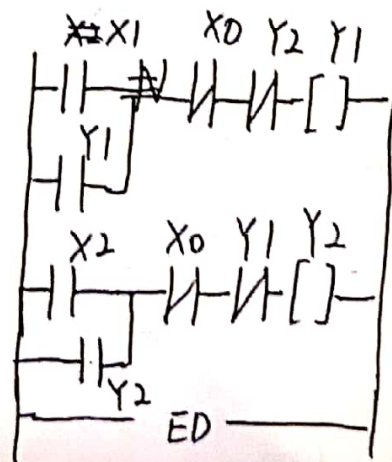
ST X2

OR Y1

ANI X1

OT Y1

ED



ST X1

OR Y1

ANI X0

ANI Y2

OT Y1

ST

≡

ED

☆ 三相对称电路计算

1) 线电压 $\dot{U}_{12} = 380 \angle 0^\circ \text{ V}$, 相电压为 $220 \angle -30^\circ \text{ V}$.

对星形负载 $\dot{I}_{1Y} = \frac{\dot{U}_1}{Z_1} = \frac{220 \angle -30^\circ}{44} = 5 \angle -30^\circ \text{ A}$

对三角形负载 其相电流 $\dot{I}_{1\Delta} = \frac{\dot{U}_{12\Delta}}{Z_2} = \frac{380 \angle 0^\circ}{72} = 5.28 \angle 0^\circ \text{ A}$

$\Rightarrow$ 线电流 $\dot{I}_{1\Delta} = \sqrt{3} \times 5.28 \angle 0^\circ - 30^\circ = 9.14 \angle -30^\circ \text{ A}$

$\dot{I}_1 = \dot{I}_{1Y} + \dot{I}_{1\Delta} = 14.14 \angle -30^\circ \text{ A}$

对称得 $\dot{I}_2 = 14.14 \angle -150^\circ \text{ A}$

$\dot{I}_3 = 14.14 \angle 90^\circ \text{ A}$

2) $P = P_\Delta + P_Y = \sqrt{3} U_L I_{1\Delta} \cos \varphi_\Delta + \sqrt{3} U_L I_{1Y}$

$= \sqrt{3} \times 380 \times 9.14 \times \cos 30^\circ + \sqrt{3} \times 380 \times 5$

φ 角是相电压与相电流之间的相位差
 U_1 I_{12}

☆ 电机铭牌计算

$I_N = \frac{P_N}{\sqrt{3} U_N \cos \varphi \eta}$

$S_N = \frac{n_0 - n_N}{n_0}$

$T_N = \frac{P_N}{\omega} = \frac{9550 P_N}{n}$

自耦变压器 $\frac{U_{P1}}{U_{P2}} = \frac{N_1}{N_2} = K$

$I_{st}' = \frac{I_{st}}{K^2}$

$T_{st}' = \frac{T_{st}}{K^2}$

$P_N = \sqrt{3} I_N U_N \cos \varphi \eta$

$P = 3 I_p U_p \cos \varphi$
 $= \sqrt{3} I_L U_L \cos \varphi$

1) $\eta = \frac{P_N}{\sqrt{3} U_N I_N \cos \varphi} = \frac{30 \times 10^3}{\sqrt{3} \times 380 \times 68 \times 0.8}$

$n_0 = 1000 \text{ r/min}$

$S = \frac{n_0 - n}{n_0} \times 100\% = \frac{1000 - 980}{1000} \times 100\%$

2) $I_{st} = 7 \times I_N$
 $T_{st} = 2 T_N$

$K = \frac{U_1}{U_2} = \frac{1}{a}$

$T_{st}' = \frac{T_{st}}{K^2} = a^2 T_{st}$

△形连接 $U_{12} = U_L$
 $U_L = U_p$
 $I_1 = \sqrt{3} I_{12}$, $I_L = \sqrt{3} I_p$
且 I_L 比 I_p 滞后 30°
Y形连接 $U_L = \sqrt{3} U_p$
 $U_1 = \sqrt{3} U_{12}$
且 U_L 比 U_p 超前 30°

负载增加是指负载电流增大(电源输出电流)

$$C = \frac{P}{\omega U^2} (\tan \varphi_1 - \tan \varphi)$$

电路作用: 传输、分配和转换 → 对电信号传递、存储和处理。

电路中的基本物理量: 1. U 、电压、电动势、电功率

有电压不一定会有电流 = 电路需有闭合回路。

支路: 无分支的两端 [对于 n 节点 b 支路的平面电路, 可列出 $(n-1)$ 个独立 KCL, $(b-n+1)$ 个 KVL

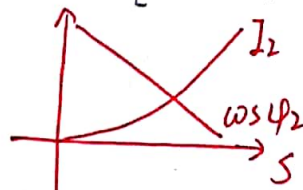
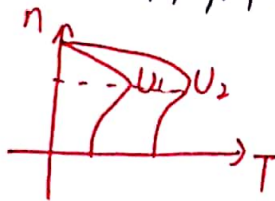
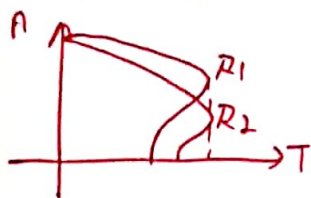
开关 S 闭合瞬间 i_R 和 i_L 发生突变 不发生

谐振, $X_C = X_L$

无功功率 $Q = \sqrt{3} U_L I_L \sin \varphi$, φ 是相电压与相电流夹角

中线是为了使不对称负载的相电压对称

变压器在 S_N 下运行时其输出有功功率的大小取决于 [负载功率因数为大]



鼠笼型属于硬特性, 绕线转子常用于起重设备

电机在运行中保险丝熔断一相, 电流大大增大

在编制 PLC 程序时, 同一继电器线圈可在程序中重复输出 (X)

OT 指令可以用于输入继电器 X (X)

单相交流电路 $P = UI \cos \varphi$

$$Q = UI \sin \varphi$$

$$S = UI$$

三相功率: $P = 3 U_p I_p \cos \varphi$

$$= \sqrt{3} U_L I_L \cos \varphi$$

8550

单相电路计算

$R \quad u=Ri \Rightarrow \dot{U} \rightarrow j \quad \varphi=0 \quad I=\frac{U}{R} \quad j=\frac{\dot{U}}{R}$

$L \quad u=L\frac{di}{dt} \Rightarrow \dot{U} \rightarrow j \quad \varphi=90^\circ \quad I=\frac{U}{X_L} \quad j=\frac{\dot{U}}{jX_L} \quad i=\frac{\dot{U}}{j\omega L}$

$C \quad u=\frac{1}{C}\frac{di}{dt} \Rightarrow \dot{U} \rightarrow j \quad \varphi=-90^\circ \quad I=\frac{U}{X_C} \quad j=\frac{\dot{U}}{-jX_C} \quad i=\frac{\dot{U}}{j\frac{1}{\omega C}}$

$R, L \text{ 串联} \Rightarrow \dot{U} \rightarrow j \quad \varphi>0 \quad I=\frac{U}{\sqrt{R^2+X_L^2}} \quad j=\frac{\dot{U}}{R+jX_L}$

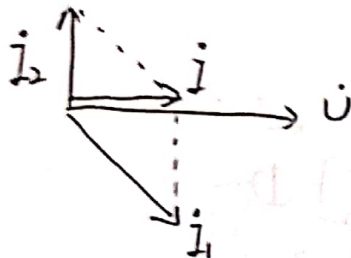
$R, C \text{ 串联} \Rightarrow \dot{U} \rightarrow j \quad \varphi<0 \quad I=\frac{U}{\sqrt{R^2+X_C^2}} \quad j=\frac{\dot{U}}{R-jX_C}$

$R, L, C \text{ 串联} \Rightarrow I=\frac{\dot{U}}{\sqrt{R^2+X_L^2+X_C^2}} \quad j=\frac{\dot{U}}{R-jX_C+jX_L} \quad i=\frac{\dot{U}}{R+j(X_L-X_C)}$

$j=\sqrt{-1}, j^2=-1$

4.5.10 由题意 $U=220V, I_1=\frac{220}{\sqrt{2}}=156A, I_2=\frac{11\sqrt{2}}{\sqrt{2}}=11A$

$\dot{I}=\dot{I}_1+\dot{I}_2=156\angle 45^\circ+11\angle 90^\circ=11\angle 0^\circ A$



得 $I=11A$

$X_C=\frac{U}{I_2}=\frac{220}{11}=20\Omega$

$X_C=\frac{1}{\omega C} \Rightarrow C=\frac{1}{\omega X_C}=\frac{1}{314 \times 20}=159\mu F$

$Z_1=R+X_Lj=\frac{\dot{U}}{\dot{I}_1}=\frac{220\angle 0^\circ}{156\angle 45^\circ}=10\sqrt{2}\angle 45^\circ\Omega=(10+10j)\Omega$

$\Rightarrow R=10\Omega$

$X_L=10\Omega=\omega L \Rightarrow L=\frac{X_L}{\omega}=\frac{10}{314}=31.8mH$

单相电路计算: 设 $\dot{U}_S=15\sqrt{2}\angle 0^\circ V$

$U_S=\frac{30}{\sqrt{2}}V$

$X_C=\frac{1}{\omega C}=\frac{1}{314 \times 600 \times 10^{-6}}\Omega=530.5\Omega$

$X_L=\omega L=314 \times 600 \times 10^{-3}\Omega=188.4\Omega$

$\dot{I}_1=\frac{\dot{U}}{Z_1}$

$\dot{I}_1=\frac{-jX_C}{R+jX_L-jX_C}$

P135

分流公式

单相电路计算也可用分流公式

$\dot{I}_1=\frac{\dot{U}_S}{R_1+j(X_L-X_C)}$

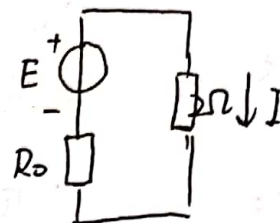
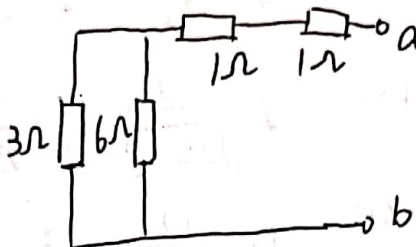
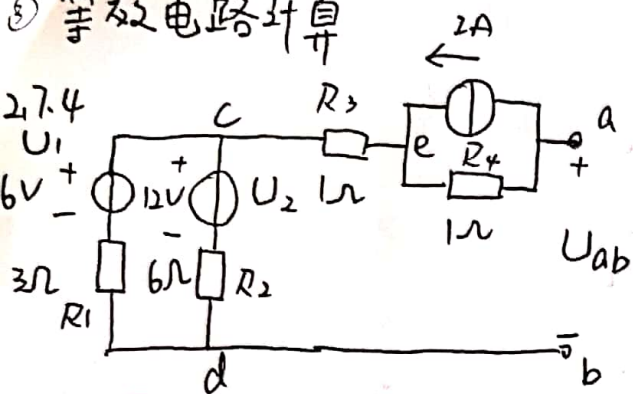
$\dot{U}_C=\dot{U}_S-\dot{I}_1 R_1$

戴维南定理, 叠加定律计算

① 开路电路计算 开路电压

② 除源电路等效内阻

③ 等效电路计算



开路电压 U_{ab} 开路

$$U_{ab} = U_{cd} + U_{ce} - U_{ea}$$

$$U_{ce} = 0$$

$$U_{ea} = I_s R_4 = 2 \times 1 = 2V$$

$$U_{cd} = \frac{\frac{U_1}{R_1} + \frac{U_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{\frac{2+2}{3+6}}{\frac{1}{3} + \frac{1}{6}} = 8V$$

$$\Rightarrow E = U_{ab} = 6V$$

$$R_o = 1 + 1 + \frac{3 \times 6}{3 + 6} = 4\Omega$$

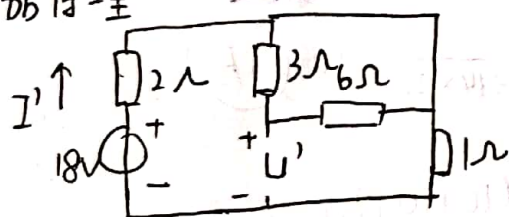
$$I = \frac{E}{R_o + R}$$

$$= \frac{6}{4 + 2}$$

$$= 1A$$

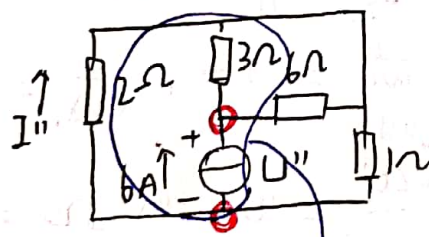
电流源断开
电压源短路

叠加原理:



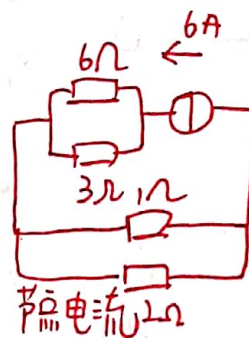
$$I' = \frac{18}{2 + 1} = 6A$$

$$U' = 6 \times 1 = 6V$$



$$I'' = \frac{1}{2 + 1} \times 6 = 2A$$

$$U'' = 6 \times \frac{3 \times 6}{3 + 6} + 2 \times 2 = 16V$$

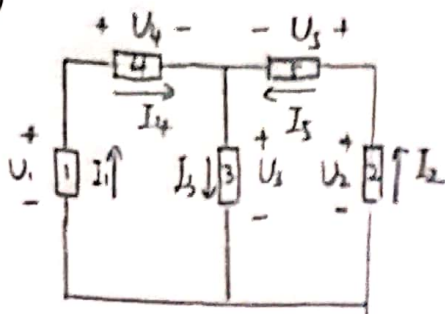


$$I = I' + I'' = 6 - 2 = 4A$$

$$U = U' + U'' = 16 + 6 = 22V$$

1.5.9

11)



12) 如果 U, I 实际方向相同肯定是负载, 方向相反是电源

1.2 是电源

3.4.5 是负载

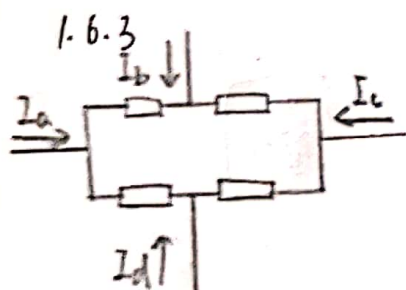
13) $P_1 = U_1 I_1 = -560 \text{ W}$ $P_2 = U_2 I_2 = -340 \text{ W}$ $P_3 = U_3 I_3 = 600 \text{ W}$

$P_4 = U_4 I_4 = 320 \text{ W}$ $P_5 = U_5 I_5 = 180 \text{ W}$

$|P_1 + P_2| = 1100 \text{ W}$

$|P_3 + P_4 + P_5| = 1100 \text{ W}$

电源发出功率和负载使用功率平衡。

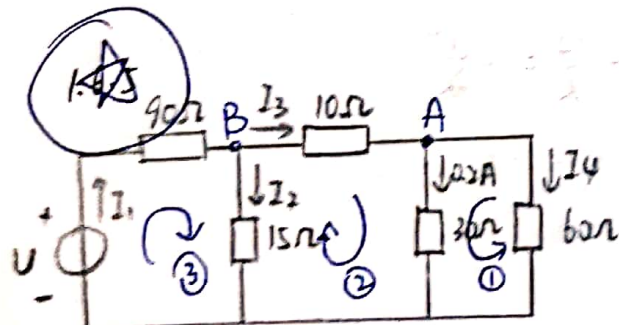


由基尔霍夫电流定律得

$$I_a + I_b + I_c + I_d = 0$$

$$1 + 10 + 2 + I_d = 0$$

$$I_d = -13 \text{ mA}$$



KVL

由基尔霍夫电压定律得

$$\textcircled{1} 0.2 \times 30 - 60 \cdot I_4 = 0$$

$$I_4 = 0.1 \text{ A}$$

$$\textcircled{2} 0.2 \times 30 - 15 \cdot I_2 + 0.3 \times 10 = 0 \quad B: I_1 - 0.3 - 0.6 = 0$$

$$I_2 = 0.6 \text{ A}$$

$$I_1 = 0.9 \text{ A}$$

$$\textcircled{3} U - 15 \times 0.6 - 90 \times 0.9 = 0$$

$$U = 17.1 \text{ V}$$

KCL

由KCL定律得

$$A: I_3 - 0.2 - 0.1 = 0$$

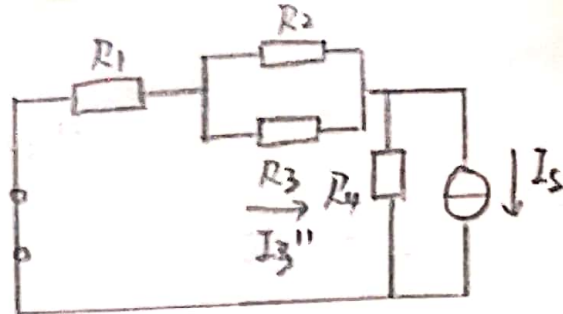
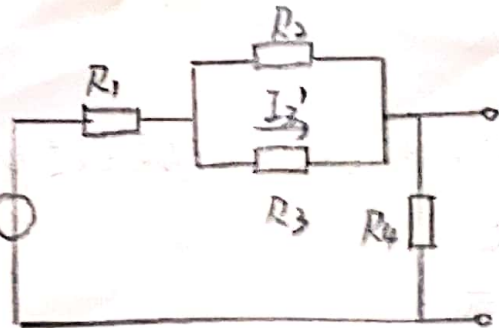
$$I_3 = 0.3 \text{ A}$$

机设183

180329

黎萧琪

2.6.3
叠加定理



$$I_3' = \frac{R_3}{R_2 + R_3} \left(\frac{U_1}{R_1 + R_4 + R_2 \parallel R_3} \right) = \frac{1}{1+1} \times \left(\frac{1}{1+1 + \frac{1}{1+1}} \right) = 0.2 \text{ A}$$

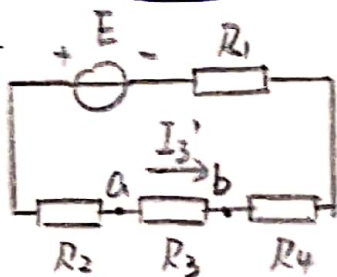
$$I_3'' = \frac{R_1 + R_2 \parallel R_3}{R_4 + R_1 + R_2 \parallel R_3} \cdot I_s \cdot \frac{R_3}{R_2 + R_3}$$

分压公式

$$I_3'' = \frac{R_4}{R_1 + R_4 + R_2 \parallel R_3} \cdot I_s \cdot \left(\frac{R_2}{R_2 + R_3} \right) = 0.4 \text{ A}$$

$$I_3 = I_3' + I_3'' = 0.2 + 0.4 = 0.6 \text{ A}$$

2.6.4



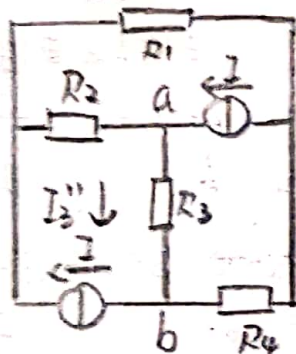
$$I_3' = \frac{E}{R_1 + R_2 + R_3 + R_4} = \frac{E}{4R}$$

$$I_3 = \frac{U_{ab}}{R_3} = \frac{U_{ab}}{R}$$

$$I_3'' = I_3 - I_3' = \frac{10}{4R} - \frac{12}{R} = -\frac{19}{2R}$$

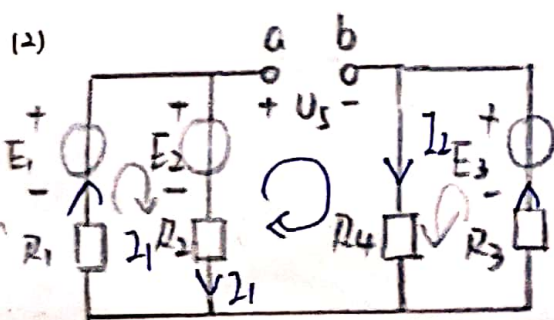
$$U_{ab}' = I_3'' \cdot R_3 = -\frac{19}{2} \text{ V} = -9.5 \text{ V}$$

8
叠加

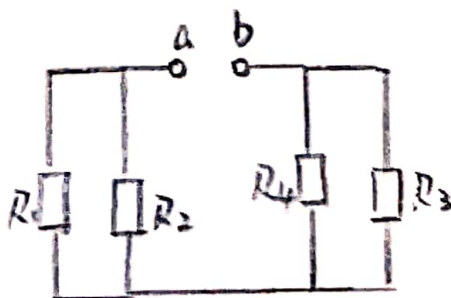


2.7.6
戴维南定理

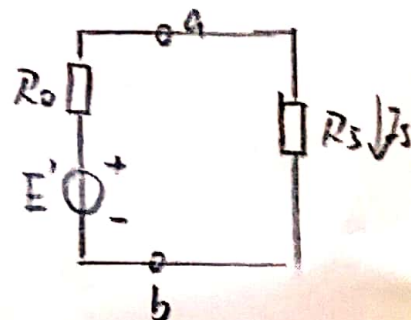
当S断开时， I_5 有闭合回路，则 $I_5 = 0$ ， $U_5 = I_5 R_5 = 0$



开路电路
 $E = U_5$



恒电路
 R_0



等效电路

$$I_1 = \frac{E_1 - E_2}{R_1 + R_2} = \frac{15 - 13}{1 + 1} = 1 \text{ A}$$

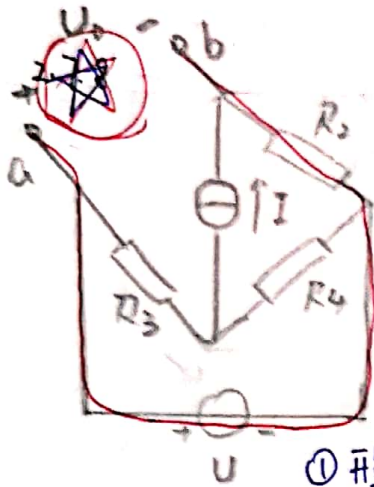
$$I_2 = \frac{E_3}{R_3 + R_4} = \frac{4}{1 + 1} = 2 \text{ A}$$

KVL:

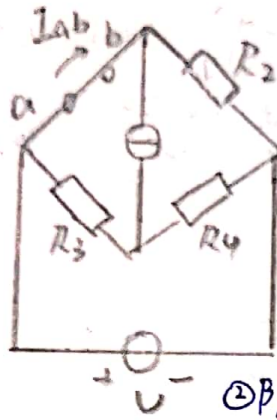
$$E = U_{ab} = I_1 R_2 + E_2 - I_2 R_4 = 1 + 13 - 2 \times 1 = 12 \text{ V}$$

$$R_0 = R_1 // R_2 + R_3 // R_4 = 1 \Omega$$

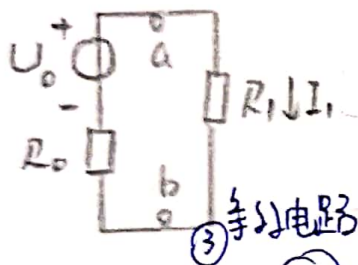
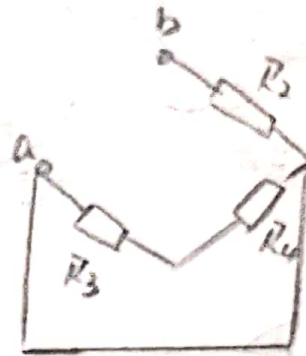
$$I_3 = \frac{U_{ab}}{R_0 + R_5} = \frac{12}{1 + 10} = 1.09 \text{ A}$$



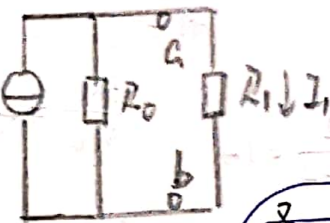
① 开路电压



② 除源电阻R0



③ 短路电路



☆ 将 a, b 短接

诺顿定理

诺顿定理

$$I_{ab} = \frac{U}{R_2} - I = \frac{10}{4} - 2 = 0.5 \text{ A}$$

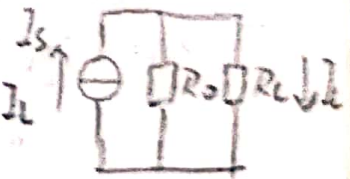
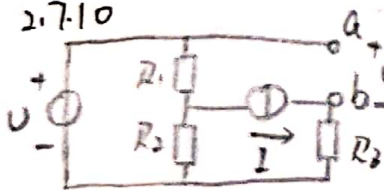
$$U_{ab} = 10 - U - 2R_2 = 10 - 2 \times 4 = 2 \text{ V}$$

$$R_0 = R_2 = 4 \Omega$$

$$I_1 = \frac{U_{ab}}{R_0 + R_1} = \frac{2}{4 + 9} = 0.154 \text{ A}$$

$$I_1 = \frac{R_0}{R_0 + R_1} I_{ab} = \frac{4}{4 + 9} \times 0.5 = 0.154 \text{ A}$$

2.7.10



$$U_{ab} = U - R_3 I = 32 - 8 \times 2 = 16 \text{ V}$$

$$R_0 = R_3 = 8 \Omega$$

$$I_L = \frac{U_{ab}}{R_0} = \frac{16}{8} = 2 \text{ A}$$

$$I_{ab} = \frac{U}{R_3} - I = \frac{32}{8} - 2 = 2 \text{ A}$$

$$I_L = \frac{R_0}{R_L + R_0} \cdot 2 I_{ab} = \frac{8}{24 + 8} \times 2 \text{ A} = 0.5 \text{ A}$$

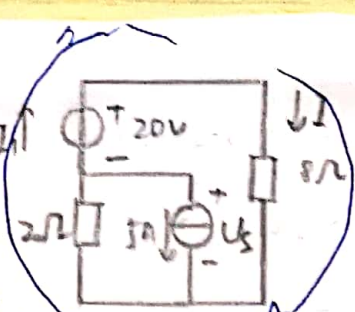
2.3.6

电源等效变换

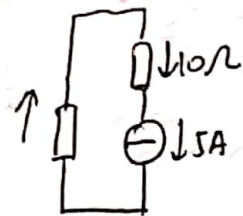
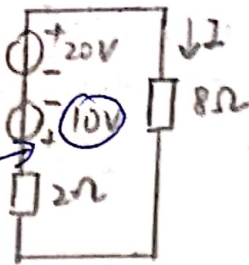
与电压源串联和
与电流源并联的
电阻保留

$$I = \frac{20-10}{2+8} = 1A$$

都是电源



$$E = I_s R_0$$



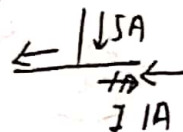
$$I_1 = \frac{20}{10} + 1 = 3A$$

回路电压 KVL

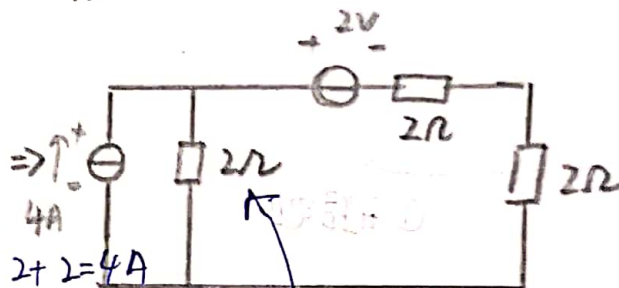
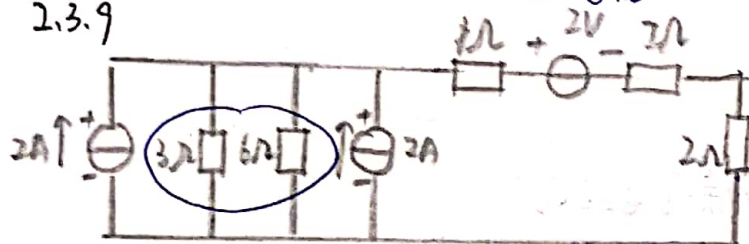
$$10 \times 5 + U_s + 2 \times (5+1) = 0$$

$$U_s = -62V$$

回路电压 KVL

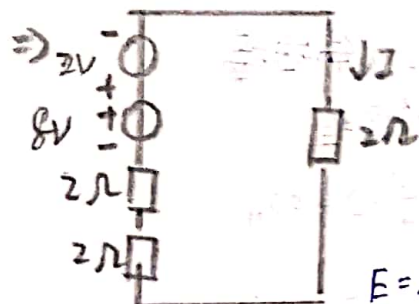


2.3.9



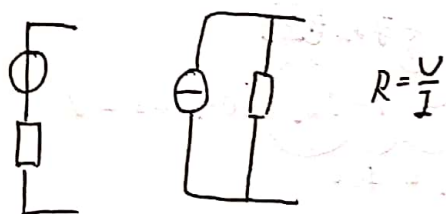
$$2+2=4A$$

$$\frac{3 \times 6}{3+6} = \frac{18}{9} = 2\Omega$$



$$I = \frac{8-2}{2+2+2} = 1A$$

$$E = I_s R = 2 \times 4 = 8V$$



$$R = \frac{U}{I}$$

2.4.2 利用结点电压法

$$U_{ab} = \frac{\frac{120}{0.8} + \frac{116}{0.4} + 10}{\frac{1}{2.8} + \frac{1}{0.4} + \frac{1}{4}} = 112.5V \quad I_1 = \frac{120 - U_{ab}}{0.8} = 9.38A$$

$$I_2 = \frac{116 - U_{ab}}{0.4} = 8.75A \quad I = \frac{U_{ab}}{R_L} = 28.1A$$

$$P_{E1} = 120 \times I_1 - I_1^2 R_{01} = 1055W$$

$$P_{E2} = 116 \times I_2 - I_2^2 R_{02} = 984W$$

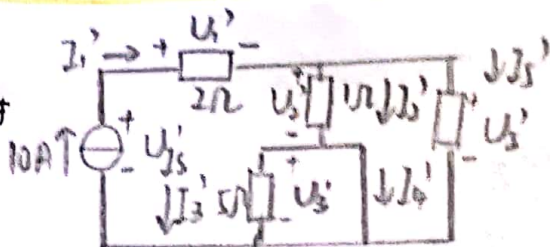
$$P_{I_s} = U_{ab} \times I_s = 112.5W$$

$$P_{R_L} = I^2 R_L = 28.1^2 \times 4 = 3164W$$

$$U = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}}$$

2.6.5

10A电源单独作用时

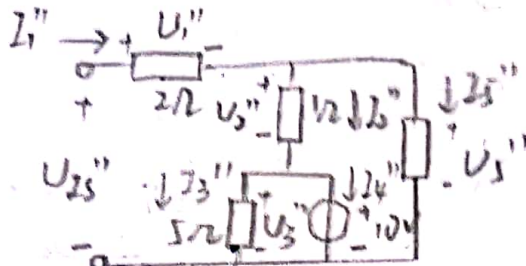


$$I_1' = 10A \quad I_2' = I_4' = \frac{4}{4+5} \times 10 = 8A \quad I_3' = 0$$

$$I_5' = \frac{1}{4+5} \times 10 = 2A \quad U_2' = 1 \times 8 = 8V \quad U_1' = 0$$

$$U_5' = 4 \times 2 = 8V \quad U_{25}' = U_1' + U_2' = 8V$$

电压源单独作用时



$$I_1'' = 0 \quad I_2'' = -\frac{10}{4+5} = -2A \quad I_3'' = \frac{10}{5} = 2A$$

$$I_4'' = -2A \quad I_5'' = \frac{10}{4+5} = 2A \quad U_1'' = 0$$

$$U_2'' = -2V \quad U_3'' = 10V \quad U_5'' = 4 \times 2 = 8V$$

$$U_{25}'' = U_5'' = 8V$$

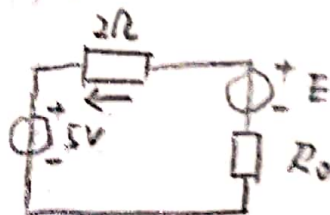
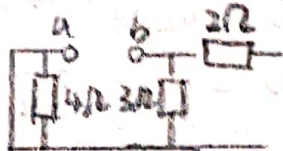
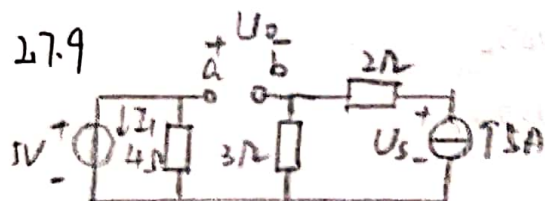
$$I_1 = I_1' + I_1'' = 10A, \quad I_2 = I_2' + I_2'' = 6A, \quad I_3 = I_3' + I_3'' = 2A$$

$$I_4 = I_4' + I_4'' = 4A, \quad I_5 = I_5' + I_5'' = 4A$$

$$U_1 = U_1' + U_1'' = 20V, \quad U_2 = U_2' + U_2'' = 8 - 2 = 6V, \quad U_3 = U_3' + U_3'' = 10V, \quad U_5 = U_5' + U_5'' = 16V, \quad U_6 = U_6' + U_6'' = 36V$$

$$\Sigma P = -10 \times 36 + 4 \times 10 + 10 \times 20 + 6 \times 6 + 2 \times 10 + 4 \times 16 = 0, \text{ 验证电路平衡}$$

2.7.9



$$(1) R_0 = 3\Omega \quad E = 3 \times 5 = 15V \quad I = \frac{E-5}{R_0+2} = 2A$$

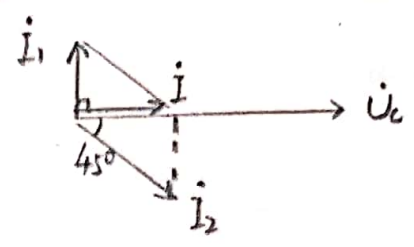
$$(2) I_1 = I \cdot \frac{5}{4} = 0.75A$$

$$U_5 = 2 \times 5 + 2 \times 2 + 5 = 19V$$

$$P_0 = 5I_1 = 3 \times 0.75 = 2.25W \text{ 取用}$$

$$P_5 = -U_5 \times I = -19 \times 2 = -38W \text{ 发出}$$

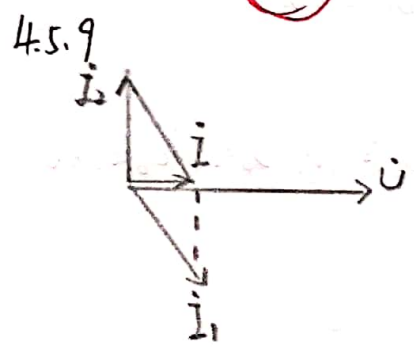
4.5.10 $\dot{U}_C = U_C \angle 0^\circ$, 则 $\dot{I}_1 = 10 \angle 90^\circ \text{ A}$
 $\varphi_1 = \arctan \frac{X_L}{R} = 45^\circ$
 $\dot{I}_2 = 10\sqrt{2} \angle -45^\circ \text{ A}$



$\dot{I} = \dot{I}_1 + \dot{I}_2 = 10 \angle 90^\circ + 10\sqrt{2} \angle -45^\circ = 10 \angle 0^\circ \text{ A}$
 $\dot{U}_C = \dot{U} - \dot{I}R$
 $\dot{U} = \dot{U}_C + \dot{I}R = 200 \angle 0^\circ - 5 \times 10 \angle 0^\circ = 150 \angle 0^\circ \text{ V}$
 $X_C = \frac{U_C}{I_1} = \frac{150}{10} = 15 \Omega$

$\dot{I} = \dot{I}_1 + \dot{I}_2$
 $X_C = \frac{U_C}{I_1}$
 $R_2 = X_L = \frac{1}{\sqrt{2}} \frac{U_C}{I_2}$

★ $R_2 = X_L = \frac{1}{\sqrt{2}} \cdot \frac{U_C}{I_2} = 7.5 \Omega$

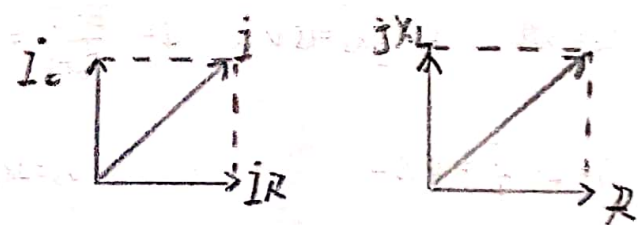


$Z_1 = \frac{\dot{U}}{\dot{I}_1} = \frac{220 \angle 0^\circ}{\frac{22}{\sqrt{2}} \angle 45^\circ} = 10\sqrt{2} \angle 45^\circ \Omega$

$R = X_L = 10 \Omega$
 $L = \frac{X_L}{2\pi f} = 0.0318 \text{ H}$

$Z_2 = \frac{\dot{U}}{\dot{I}_2} = \frac{220 \angle 0^\circ}{11 \angle 90^\circ} = 20 \angle -90^\circ \Omega$
 $X_C = 20 \Omega$
 $C = \frac{1}{2\pi f X_C} = 159 \mu\text{F}$

4.5.14
 $\dot{I}_C = \frac{\dot{U}_C}{-jX_C} = \frac{1 \angle 0^\circ}{2 \angle -90^\circ} = 0.5 \angle 90^\circ \text{ A}$
 $\dot{I}_R = \frac{\dot{U}_C}{R} = \frac{1 \angle 0^\circ}{2} = 0.5 \text{ A}$



$\dot{I} = \dot{I}_C + \dot{I}_R = 0.5 \angle 90^\circ + 0.5 = 0.5\sqrt{2} \angle 45^\circ \text{ A}$
 $\dot{U} = \dot{I}(2 + j2) + \dot{U}_C = 0.5\sqrt{2} \angle 45^\circ \times 2\sqrt{2} \angle 45^\circ + 1 \angle 0^\circ = \sqrt{2} \angle 63.4^\circ \text{ V}$

$\boxed{jX_L \quad -jX_C}$

4.5.11 $\dot{U} = 220 \angle 0^\circ \text{ V}$
 $Z_1 = 10 + j10\sqrt{3} = 20 \angle 60^\circ \Omega$
 $\dot{I}_1 = \frac{\dot{U}}{Z_1} = \frac{220 \angle 0^\circ}{20 \angle 60^\circ} = 11 \angle -60^\circ \text{ A}$
 $\dot{I}_2 = \frac{\dot{U}}{R_2} = 11 \angle 0^\circ \text{ A}$
 $\dot{I} = \dot{I}_1 + \dot{I}_2 = 19.05 \angle -30^\circ \text{ A}$

★ φ 为电流与电压夹角
 $P = UI \cos \varphi$
 $P = UI \cos (-30^\circ) = 3630 \text{ W}$

102
45.10
48.1

$$P = \frac{1}{3} \times 180 = 60 \text{ W}$$

$$I_1 = \frac{P}{U_1} = \frac{60}{220 \angle 0^\circ} = 0.273 \angle 0^\circ \text{ A}$$

由对称得 $I_2 = 0.273 \angle -120^\circ \text{ A}$

$$I_3 = 0.273 \angle 120^\circ \text{ A}$$

单相电路分析 $I_3'' = \frac{P_B}{U_3 \cos \varphi} = \frac{40}{220 \angle 120^\circ \cdot 0.5} = 0.364 \angle 60^\circ \text{ A}$

由 $\cos \varphi = 0.5$ 得, I_3'' 滞后于 U_3 60° .

由 KCL 得 $I_3 = I_3' + I_3'' = 0.273 \angle 120^\circ + 0.364 \angle 60^\circ = 0.553 \angle 85.3^\circ \text{ A}$

$$I_N = I_1 + I_2 + I_3 = 0.364 \angle 60^\circ \text{ A}$$

5.3.2 级 $U_A = 220 \angle 0^\circ \text{ V}$, 则 $U_{AB} = 380 \angle 30^\circ \text{ V}$

对星形负载 $I_1 = \frac{220 \angle 0^\circ}{10} = 22 \angle 0^\circ \text{ A}$

对三角形负载 $I_2 = \frac{(\sqrt{3} \times 380 \angle 0^\circ)}{38} = 10\sqrt{3} \angle 0^\circ \text{ A}$

$$I = I_1 + I_2 = 22 \angle 0^\circ + 10\sqrt{3} \angle 0^\circ = 39.32 \text{ A}$$

5.4.1 $I_1 = \frac{P}{\sqrt{3} U_1 \cos \varphi} = \frac{11.43 \times 10^3}{\sqrt{3} \times 380 \times 0.87} = 19.96 \text{ A}$

$$I_p = \frac{I_1}{\sqrt{3}} = \frac{19.96}{\sqrt{3}} = 11.5 \text{ A}$$

$$\dot{U}_1 = 220 \angle 0^\circ \text{ V}, \dot{U}_{12} = 380 \angle 30^\circ \text{ V}$$

对星形负载, 其相电流即线电流

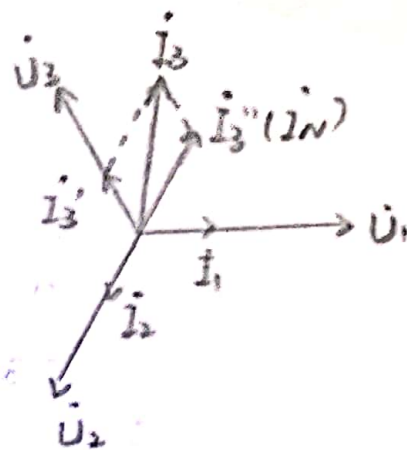
$$I_{1Y} = \frac{\dot{U}_1}{R} = \frac{220 \angle 0^\circ}{10} = 22 \angle 0^\circ \text{ A}$$

对三角形负载, 其相电流

$$I_{12\Delta} = \frac{\dot{U}_{12}}{R} = \frac{380 \angle 30^\circ}{10} = 38 \angle 30^\circ \text{ A}$$

线电流 $I_{1X} = I_{12\Delta} \cdot \sqrt{3} = 10\sqrt{3} \angle 0^\circ \text{ A}$

线电流相加 $I = I_{1\Delta} + I_{1Y}$



相线和火线之间为相电压 U_p U_1, U_2
相线之间电压为线电压 U_L U_{12}, U_{23}

$$U_L = \sqrt{3} U_p$$

每相负载上的电流称为相电流 I_p I_{12}, I_{23}

每根相线上的电流称为线电流 I_L I_1, I_2

由 Y132S-4 型号可得 $p=2$

$\Rightarrow n_1 = 1500 \text{ r/min}$

$S_N = \frac{n_1 - n}{n_1} = \frac{1500 - 1440}{1500} \times 100\% = 4\%$

$I_N = \frac{P_N}{\sqrt{3} U_N \cos \varphi_N \eta_N} = \frac{5.5 \times 10^3}{\sqrt{3} \times 380 \times 0.84 \times 0.855} = 11.64 \text{ A}$

$S_N = \frac{n_1 - n}{n_1} \times 100\%$
 $I_N = \frac{P_N}{\sqrt{3} U_N \cos \varphi_N \eta_N}$

$T_N = 9550 \frac{P_N}{n_N}$

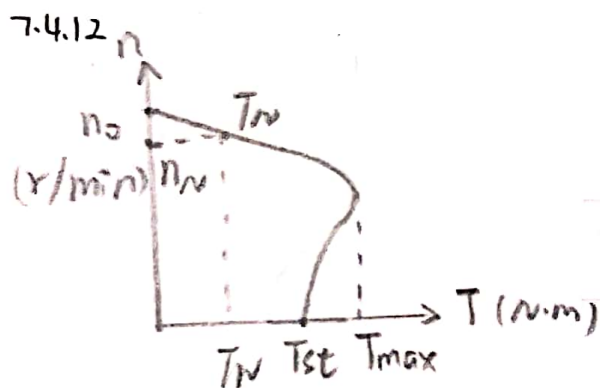
$T_N = 9550 \frac{P_N}{n_N} = 9550 \times \frac{5.5}{1440} = 36.5 \text{ N}\cdot\text{m}$

$I_{st} = 7 I_N = 81.4 \text{ A}$

$T_{st} = \frac{2}{3} T_N = 80.3 \text{ N}\cdot\text{m}$

$2.2 T_{st} = T_{max} = 80.3 \text{ N}\cdot\text{m}$

额定一线电压
线电流



$\frac{U_1^2}{T_N} = \frac{U_1^2}{T_m}$

$380 \times (1 - 5\%) = 361 \text{ V}$

$U_1 = \sqrt{\frac{T_N}{T_m}} \cdot U_1 = \sqrt{\frac{1}{2.2}} \times 380 = 256.2 \text{ V}$

$p=2 \Rightarrow n_0 = \frac{60 f_0}{p} = \frac{60 \times 50}{2} = 1500 \text{ r/min}$

(1) $n_N = (1 - S_N) n_0 = (1 - 0.04) \times 1500 = 1470 \text{ r/min}$
 $n_0 - n = S n_0 = 0.04 \times 1500 = 30 \text{ r/min}$

相对转速 $1500 - 1470 = 30 \text{ r/min}$

(2) $T_N = 9550 \times \frac{P_N}{n_N} = 9550 \times \frac{30}{1470} = 194.9 \text{ N}\cdot\text{m}$
 $T_N = 9550 \frac{P_N}{n_N}$

(3) $\cos \varphi_N = \frac{P_N}{\sqrt{3} U_L I_L \eta_N} = \frac{30 \times 10^3}{\sqrt{3} \times 380 \times 57.5 \times 0.9} = 0.88$

$P_N = \sqrt{3} U_L I_L \cos \varphi_N \eta_N$

7.5.14

$$I_{\Delta st} = 7 \times I_N = 7 \times 57.5 = 402.5 A$$

$$T_{\Delta st} = 1.2 T_N = 1.2 \times 194.9 = 233.88 N \cdot m$$

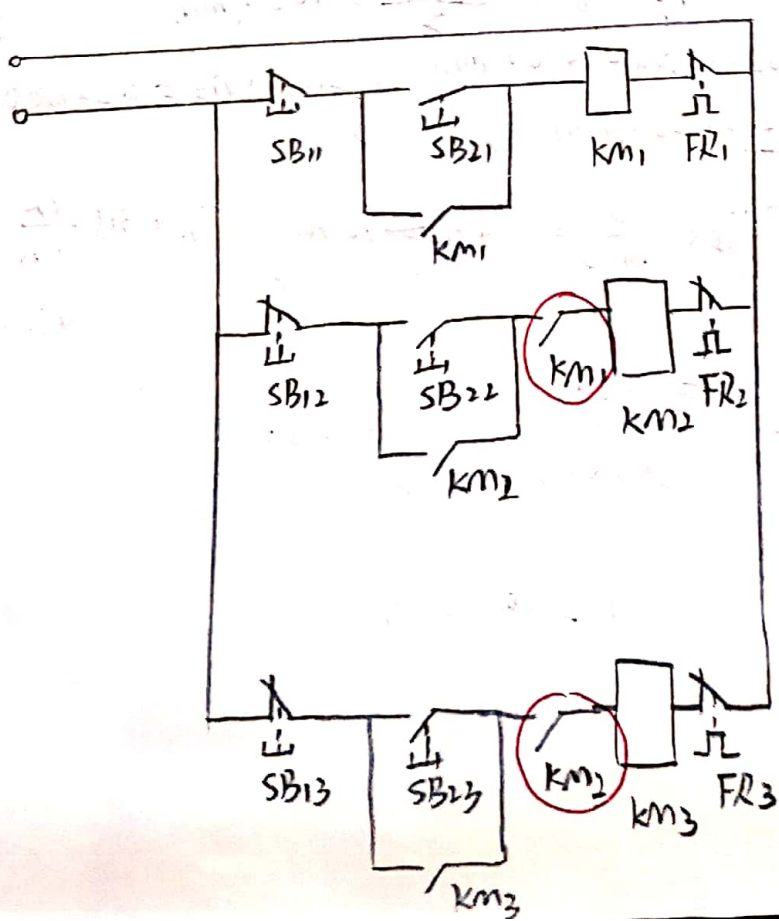
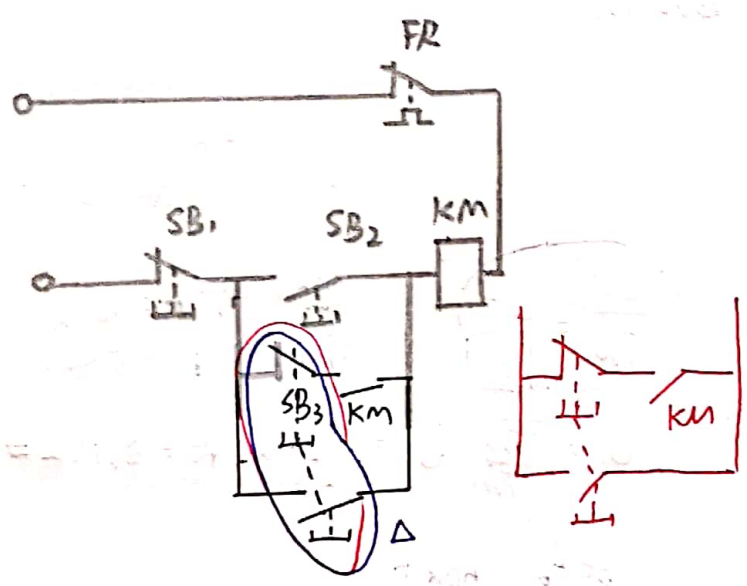
$$\Rightarrow I_{Yst} = \frac{1}{3} I_{\Delta st} = 134.2 A$$

$$T_{Yst} = \frac{1}{3} T_{\Delta st} = 77.96 N \cdot m$$

12) $T_L = 60\% \times T_N = 116.94 N \cdot m \leq T_{Yst}$, 不能启动

$T_L = 25\% \times T_N = 48.725 N \cdot m < T_{Yst}$, 能启动

10.2.3

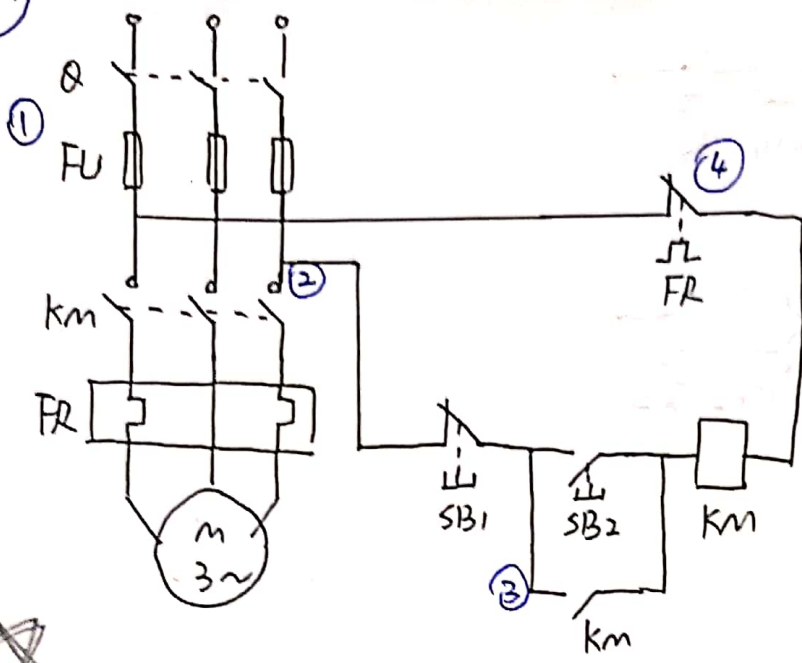


顺向启动

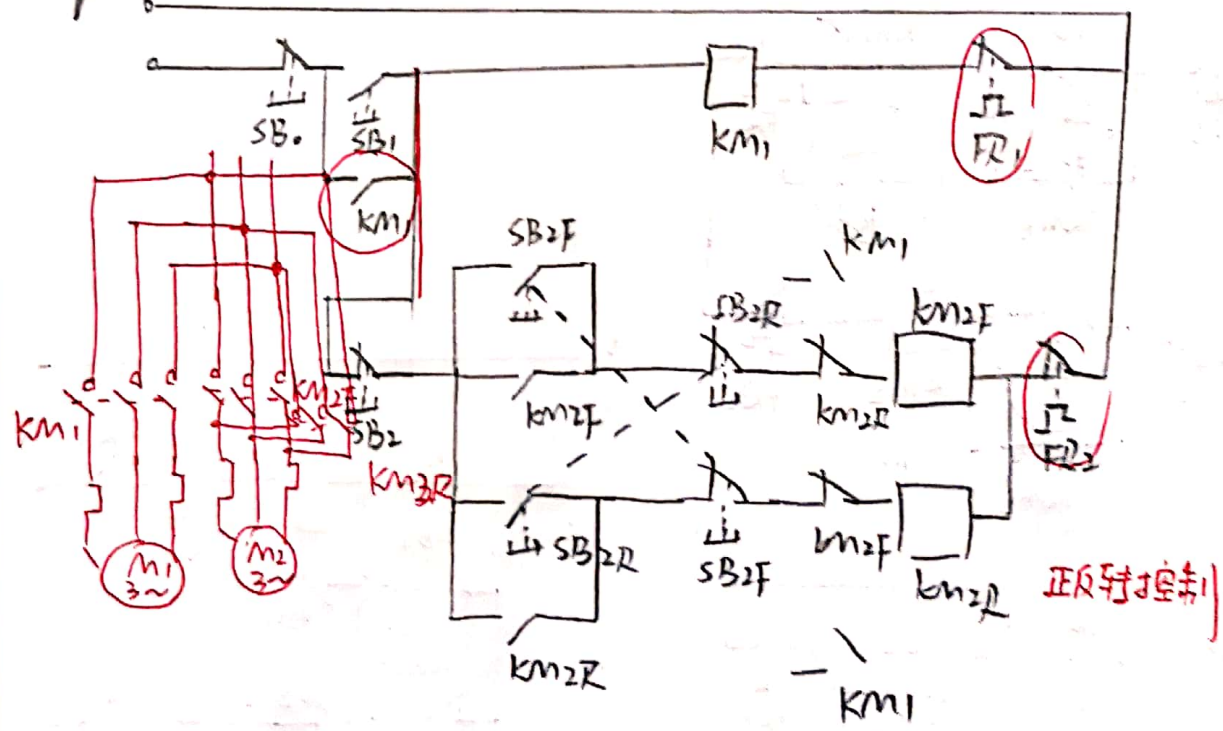
电机

10.2.7

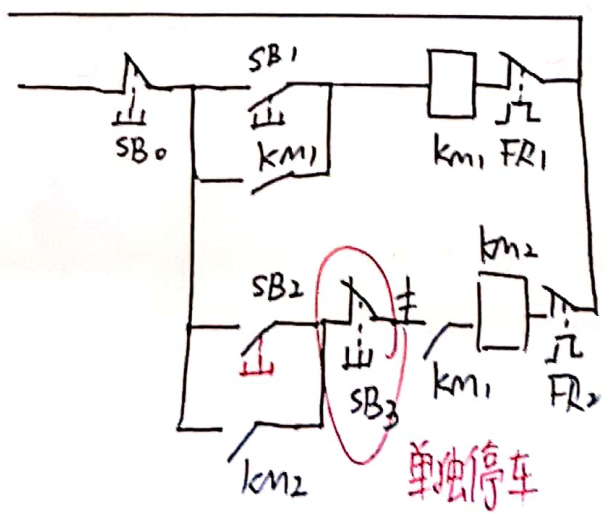
4处错误



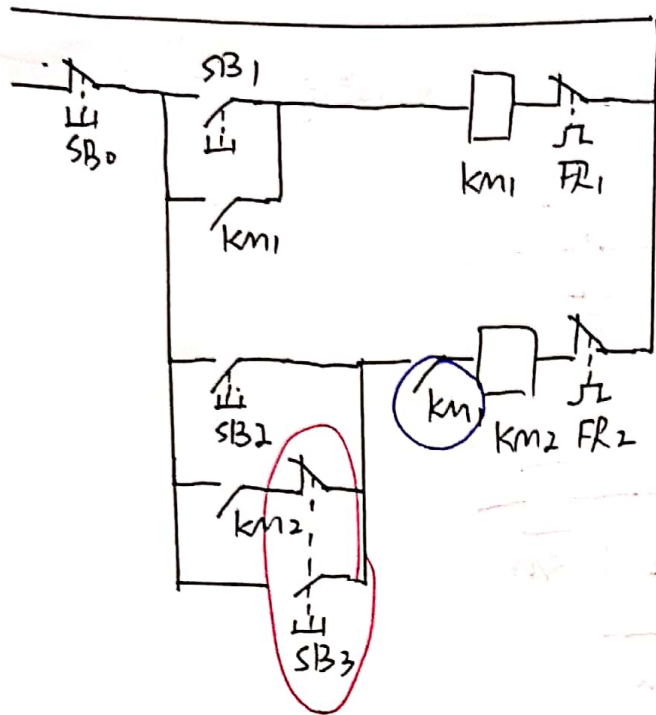
10.3.2



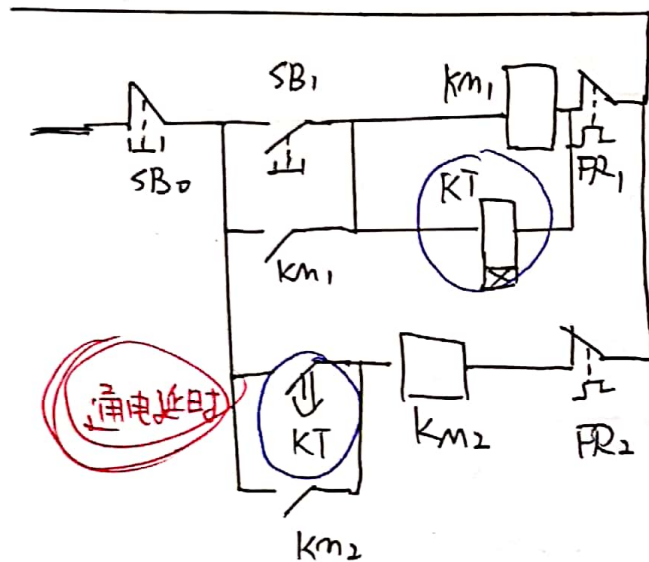
10.5.1
11)



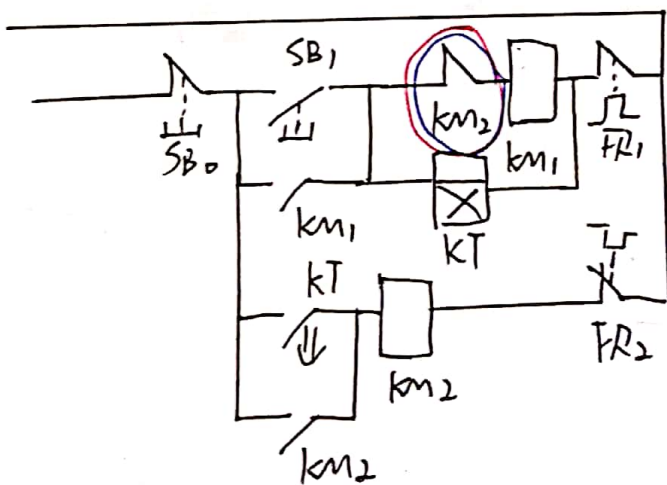
12)



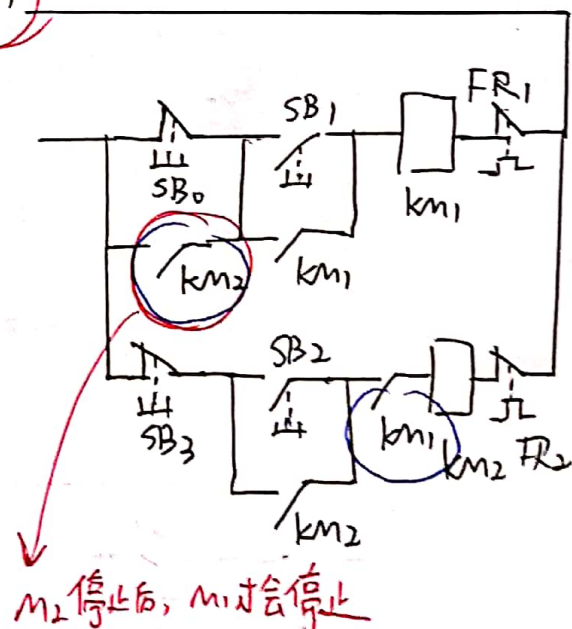
13)



14)

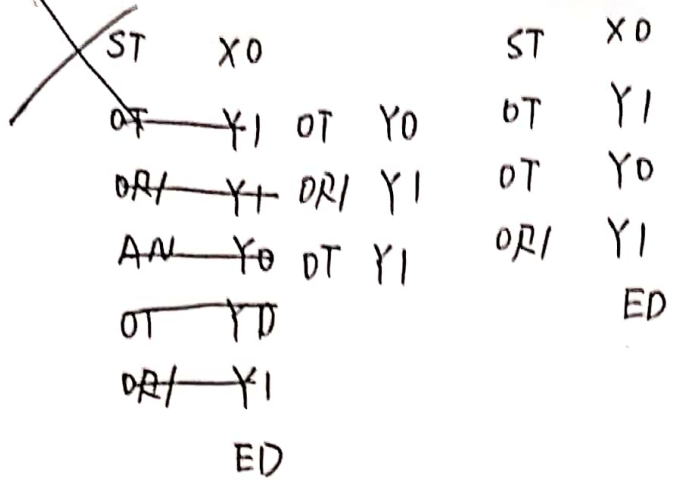


15)



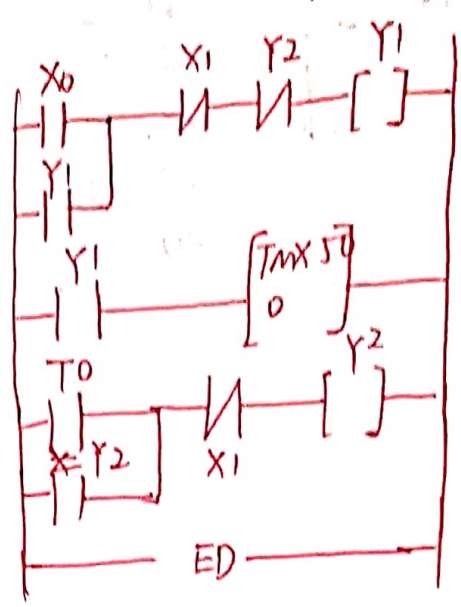
M2停止后, M1才会停止

10.2.8

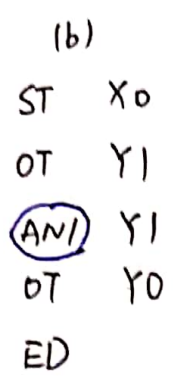
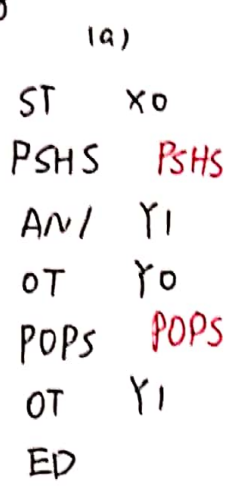


(a)

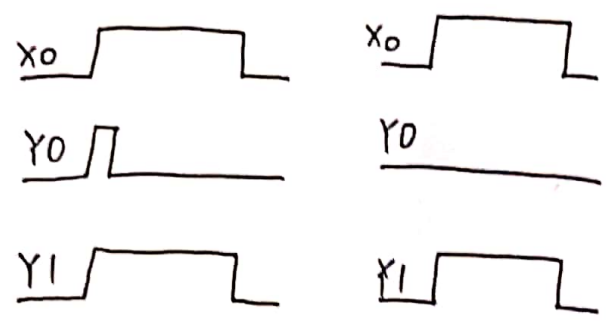
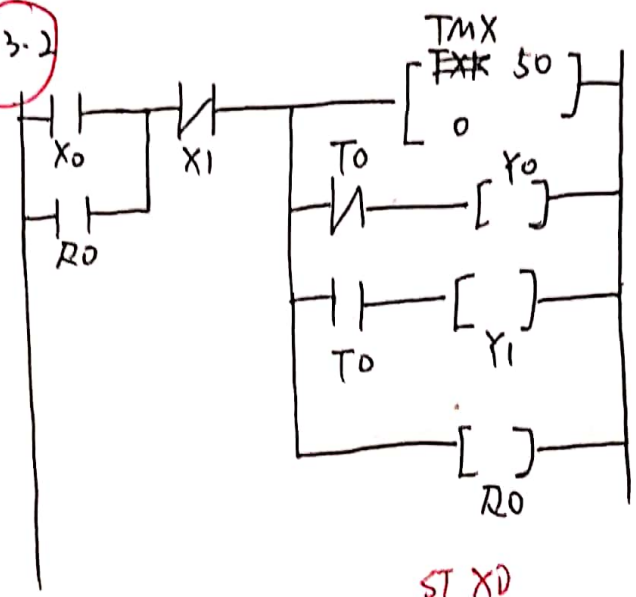
(b)



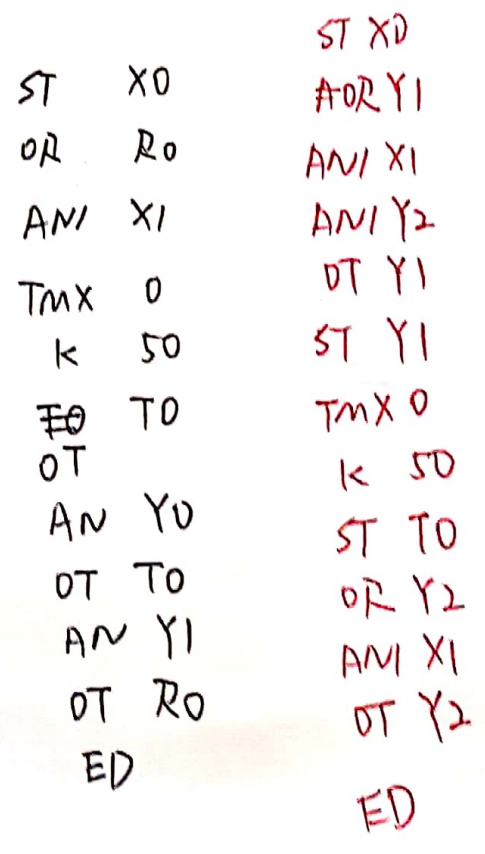
11.2.8

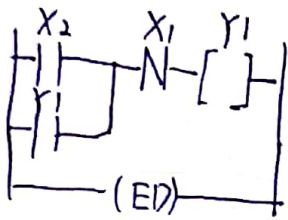


11.3.2



Y0与Y1的输出位置被颠倒,因此在(a)中的脉冲输出Y0在(b)中被输出Y1屏蔽掉了。





ST X2
 OR Y1
 AN/ X1
 OT Y1
 ED)